BACKPLANE / WIRED READING COPY

Same 137 components, 94 nets and 366 connected pins as backplane.kicad_sch.

No electronic or mechanical components were added, removed or changed.

Original hierarchical schematic and PCB retained. 20 October 2026.

HOW TO READ

Each channel groups tone, audio, microphone, analyser, LED data and photosensors.

Follow the continuous green wires. Dots join wires; crossings without dots stay separate.

Repeated A1, JA1, JVx or harness references show contacts of the same physical connector.

Pin numbers are unchanged. Contact placement is logical, not a physical connector layout.

Captions are plain text: they do not make electrical connections.

These pages crop one continuous master sheet. Shared rails continue across crop boundaries.

DRAWING CHANGES / WHY / NECESSARY?

Continuous wires replace labels: requested so connections can be followed visually.

Five channel sections and split connector symbols: readability only; no extra connectors.

Signal captions and explicit junction dots: readability only; no extra electrical function.

Cropped reading pages: easier navigation; the master PDF and KiCad sheet show full continuity.

CHECKS AND COMPONENT EXPLANATIONS

Native KiCad netlists match the original, circuit.json and PCB. ERC: zero violations.

See README.md for each existing component family: purpose and whether it is necessary.

Drawing equivalence does not establish physical hardware behavior.

BACKPLANE / CONTINUOUS-WIRE SCHEMATIC

137 original components | Actual pin numbers | No electrical net labels | 20 Oct 2026

Dots join wires. Crossings without dots do not join. Blue captions are plain text only.

Repeated connector references are contacts of the SAME physical connector. This is a logical wiring drawing.

A1 = Mega-pattern controller socket | JA1 = analyser socket | JV1-JV5 = voice sockets | J5/J1/A-J3/B-J4 = harness

GND
AGND
+5V
MEGA_5V
+12V
dxl_data
analyser/strobe
analyser/reset
amp shutdown

01 FEMALE1 / SIGNAL PATHS

CONTROLLER / SOURCE

EXISTING CONDITIONING / TEST PADS

CARD / HARNESS CONTACT

TONE

A1 D6 — TPV1 — RLT1 1K5 — DT1 YELLOW — JV1 1

FILTER OUT -> LINE OUT

JV1 3 — TP3 — TPL1 — R303 100R — J5 12

RL1 100K

AUDIO RETURN

J5 4 — JP4 0R

MICROPHONE -> 4K7 -> DIRECT INPUT

J5 6 — TPM1 — RM1 4K7 — TPD1 — A1 D26

RMD1 1M

JA1 1

ANALYSER OUTPUT

JA1 11 — TPA1 — A1 A0

NEOPIXEL

A1 D14 — RN1 330R — J5 13

RND1 100K

PHOTOSENSOR / 10K PROVISIONAL

J5 5 — A1 A5

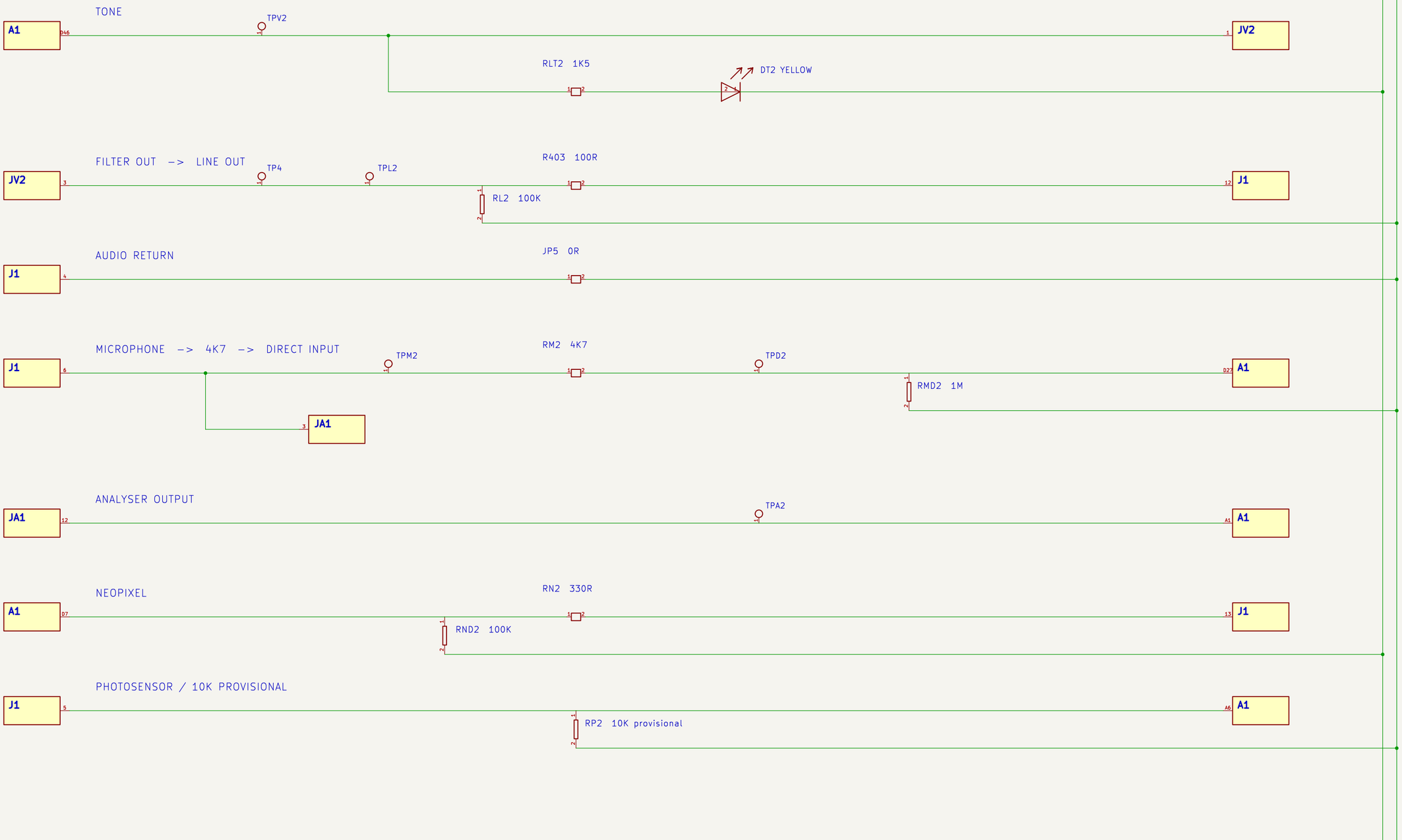
RP1 10K provisional

02 FEMALE2 / SIGNAL PATHS

CONTROLLER / SOURCE

EXISTING CONDITIONING / TEST PADS

CARD / HARNESS CONTACT

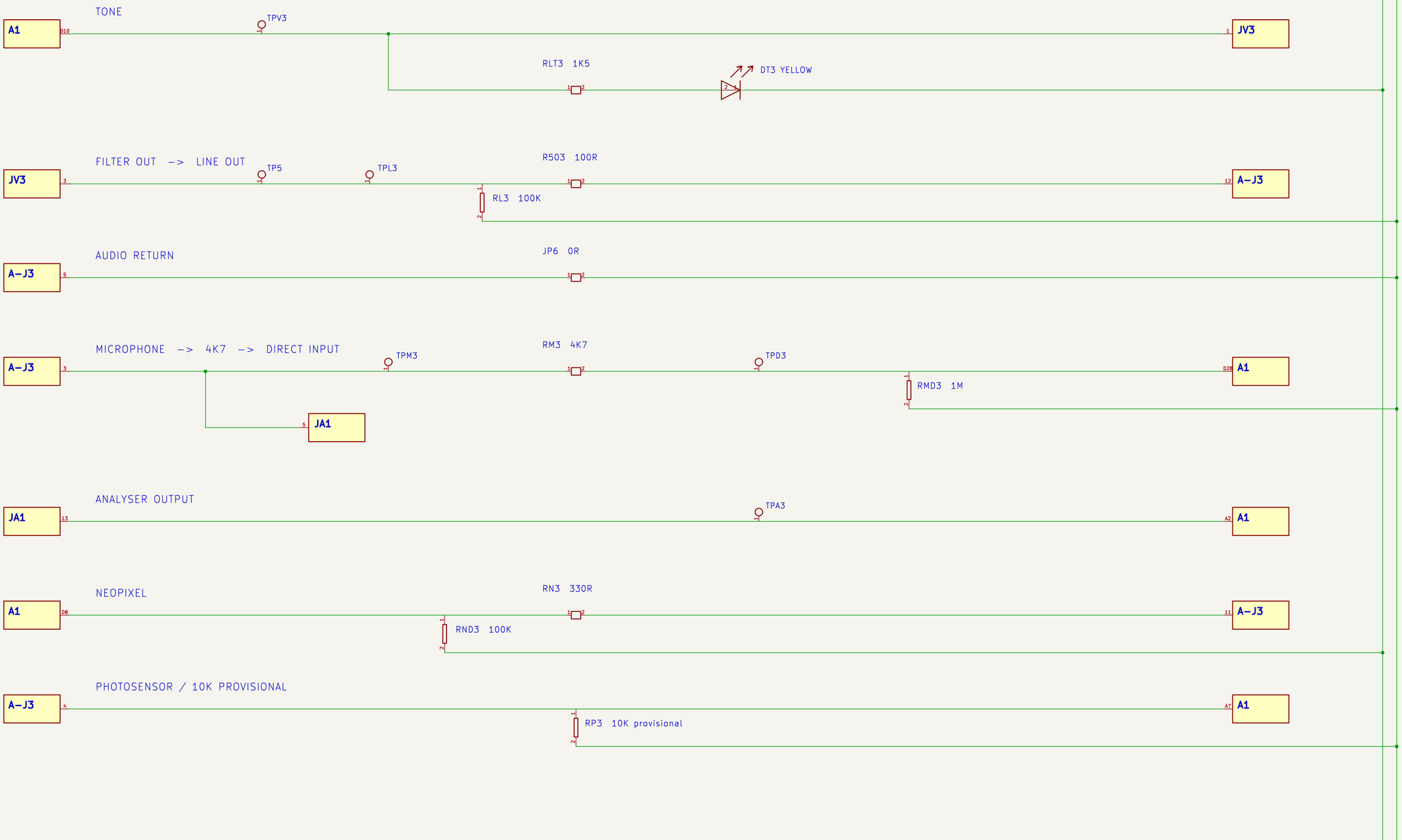


03 FEMALE3 / SIGNAL PATHS

CONTROLLER / SOURCE

EXISTING CONDITIONING / TEST PADS

CARD / HARNESS CONTACT



04 MALE1 / SIGNAL PATHS

CONTROLLER / SOURCE

EXISTING CONDITIONING / TEST PADS

CARD / HARNESS CONTACT



05 MALE2 / SIGNAL PATHS

CONTROLLER / SOURCE

EXISTING CONDITIONING / TEST PADS

CARD / HARNESS CONTACT



06 POWER / SHARED CONTROL / SPARES

C1 470uF 25V

C2 100nF

JP1 0R

RS1 10K

RLP5 2K2

DP5 GREEN

RLP12 4K7

DP12 GREEN

RLPUSB 2K2

DPUSB GREEN

analyser/strobe

analyser/reset

amp shutdown

+5V

MEGA_5V

+12V

GND

MEGA_5V

TP20

TP21

TP30

TP31

TP32

TP33

TP34

TPUSB

MEGA_5V

AGND

AGND

TPUSB

TPVG

TPEG

GND

A1 GND1

A1 GND2

A1 GND3

A1 GND4

A1 GND5

A1 GND6

J2 2

J6 1

J6 2

J6 4

GND

J7 1

J5 0

J5 8

J1 0

J1 8

A-J3 0

A-J3 1

B-J4 0

JA1 20

JA1 21

GND

JV1 2

JV2 2

JV3 2

JV4 2

JV5 2

AGND

JA1 2

JA1 4

JA1 6

JA1 8

JA1 10

JA1 16

JV1 4

JV2 4

JV3 4

JV4 4

AGND

JV5 4

+5V

J2 1

J6 3

J6 5

J6 6

J5 15

J1 15

A-J3 9

JV1 5

JV2 5

JV3 5

+5V

JV4 5

JV5 5

MEGA_5V

A1 5V1

A1 5V2

A1 5V3

A1 5V4

JA1 19

+12V

J7 2

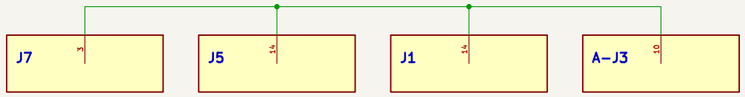
J5 7

J1 7

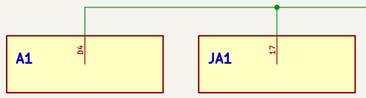
A-J3 2

dxl_data

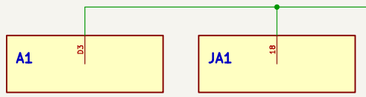
dxl_data



analyser/strobe

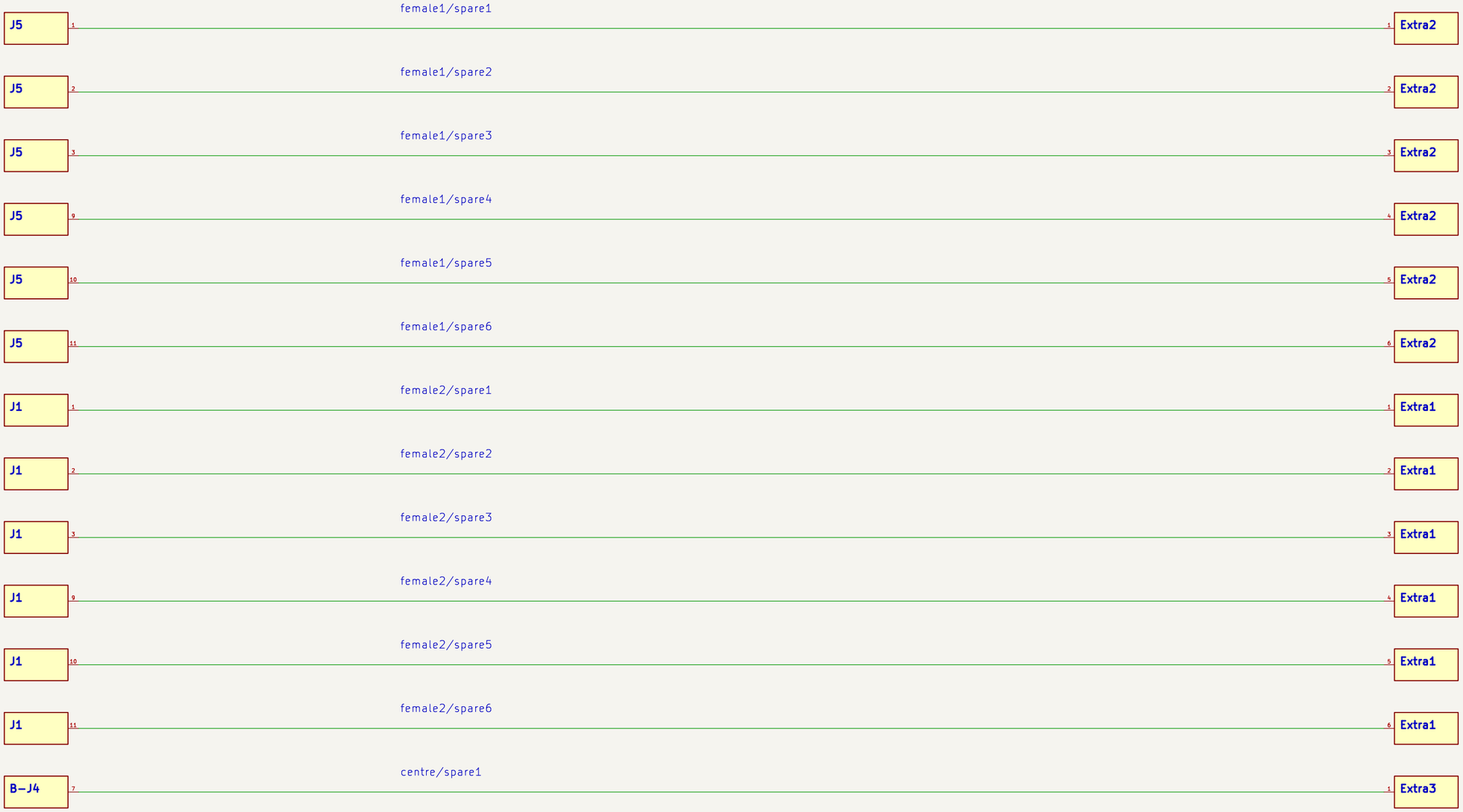


analyser/reset

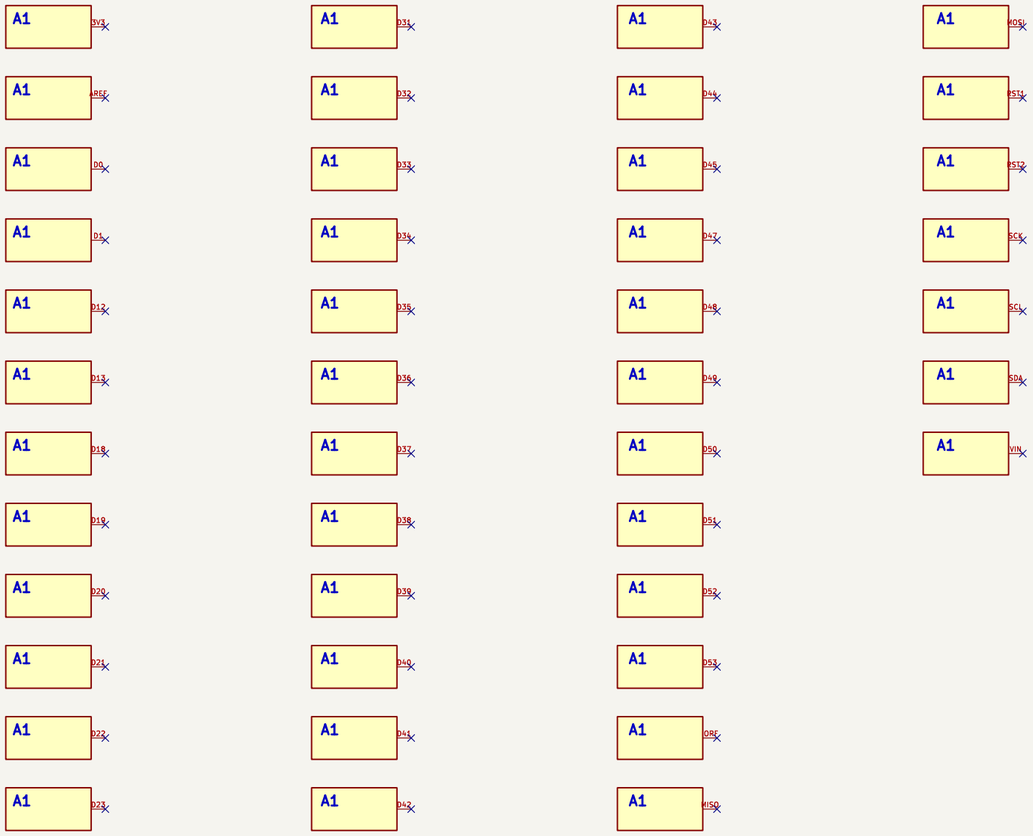


amp shutdown





07 UNUSED CONTACTS / MECHANICAL ITEMS



M1 / U2D2

HV1 / M3 card retention

HV2 / M3 card retention

HV3 / M3 card retention

HV4 / M3 card retention

HV5 / M3 card retention

HA1 / M3 analyser retention

HA2 / M3 analyser retention

amp shutdown

analyser/reset

analyser/strobe

dxl_data

+12V

MEGA_5V

+5V

AGND

GND

Reading copy: original components and connections preserved

Colloquy of Mobiles

Sheet: /

File: backplane-wired.kicad_sch

Title: Backplane / continuous wiring / reading copy